Agile Devops Lab FAT

Name:P.Sourabhi

Reg.No:23MIS0394

Set-5

Online Donation System

Problem Statement:

Ensure donations are processed and recorded accurately.

Tasks:

1.Develop donation processing logic

2.Write Junit tests for transaction validation

3.Implement CI/CD pipeline with automated testing

Set up version control using Github

Configure CI pipeline(e.g.,Jenkins/Github Actions) to trigger on code commits

Automate build process using Maven

Integrate automated unit tests in pipeline

Configure CD pipeline for automatic deployment to staging/production

4.Deploy services on Docker and Kubernetes

Containerize application using Docker (write Dockerfile)

Build and push Docker image to container registry(Docker Hub)

Create Kubernetes deployment and service YAML files

Github repo link:
maven:

1.Develop donation processing logic

Code:

```
package com.donation;

public class Donation {

   private String donorName;

   private double amount;

   public Donation(String donorName, double amount) {

      this.donorName = donorName;
```

```
        this.amount = amount;
    }
    public String getDonorName() {
        return donorName;
    }
    public double getAmount() {
        return amount;
    }
}
```

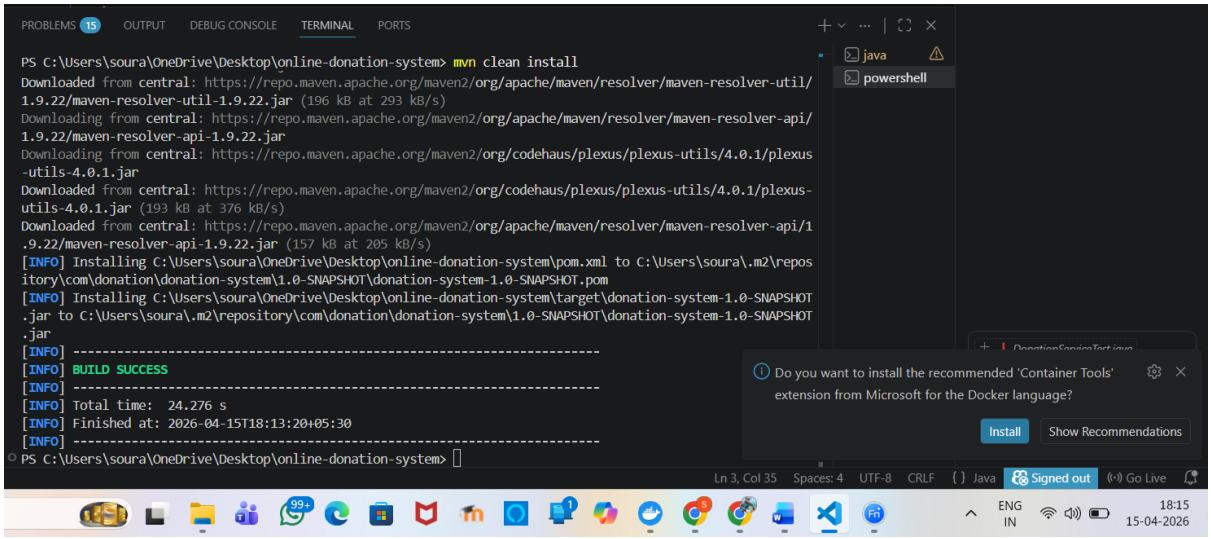
2.Write Junit tests for transaction validation

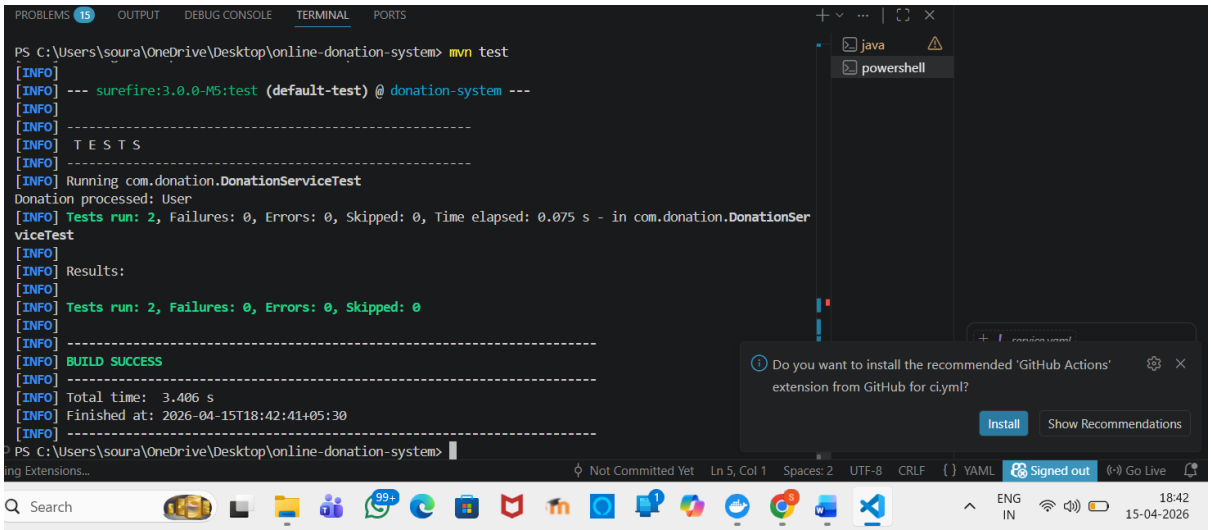
Code:

```
package com.donation;
import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;
public class DonationServiceTest {
    DonationService service = new DonationService();
    public void testValidDonation() {
        Donation d = new Donation("User", 500);
        assertTrue(service.processDonation(d));
    }
    public void testInvalidDonation() {
        Donation d = new Donation("User", -100);
        assertFalse(service.processDonation(d));
    }
}
```

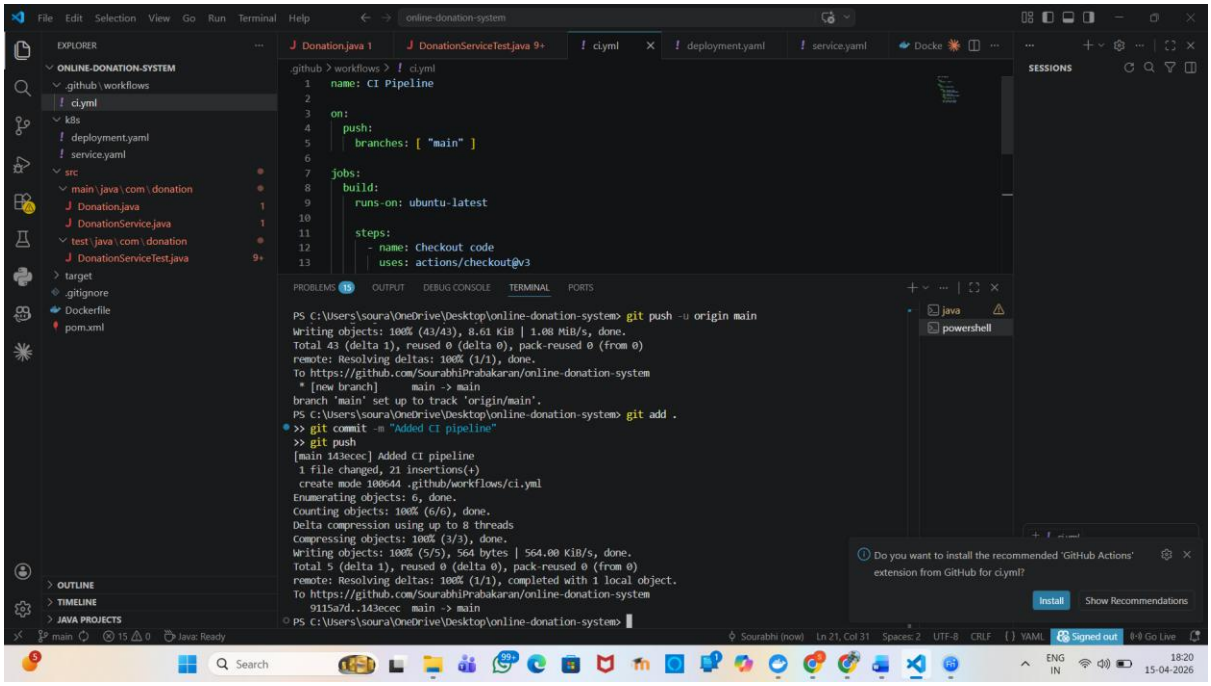
3.Implement CI/CD pipeline with automated testing

Maven:

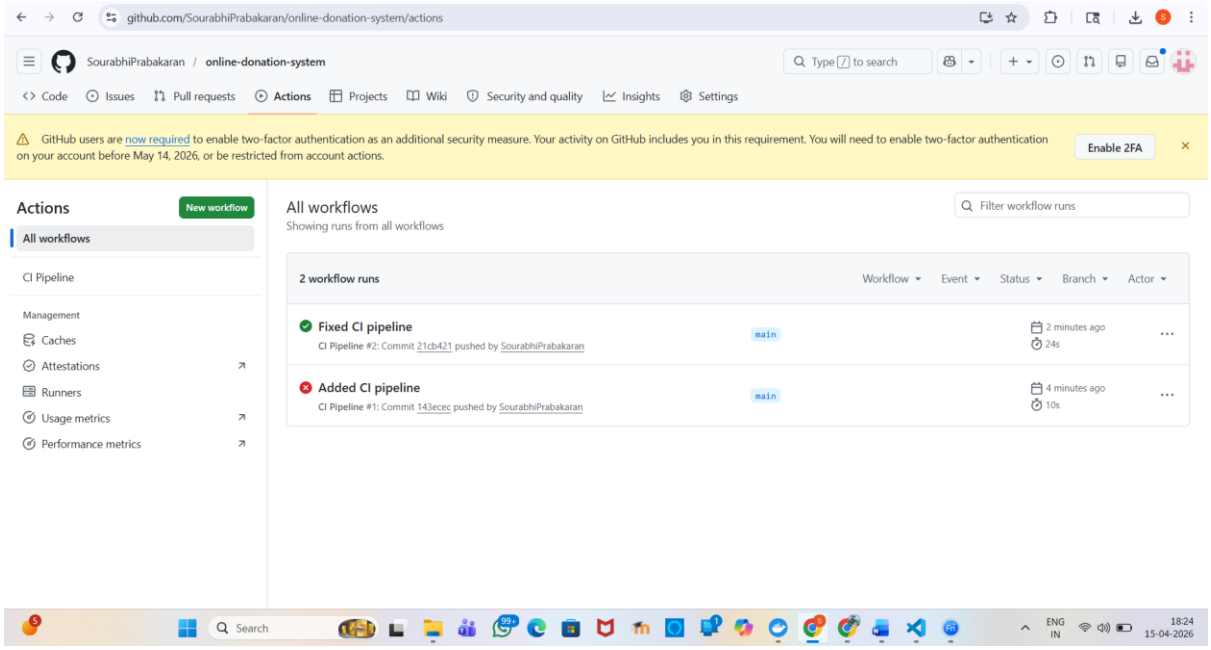




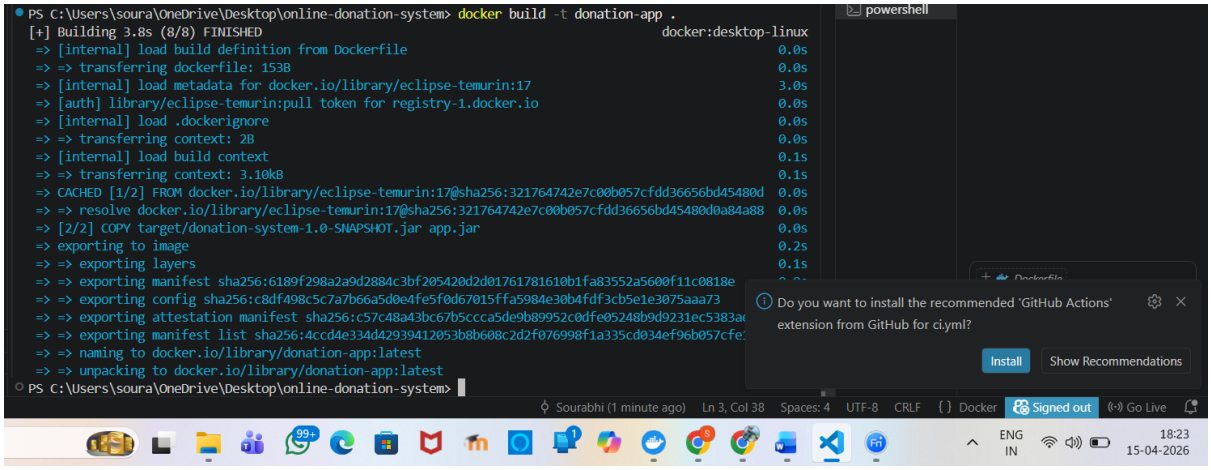
Git:

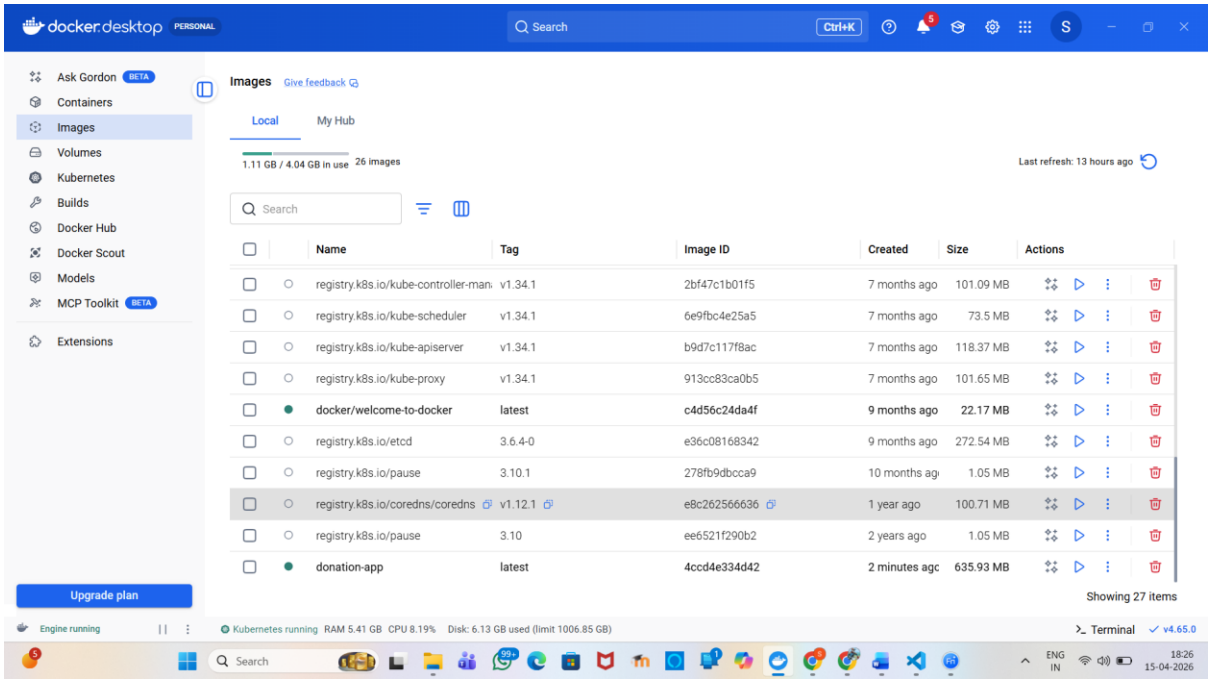


Pipeline:GitHub Actions



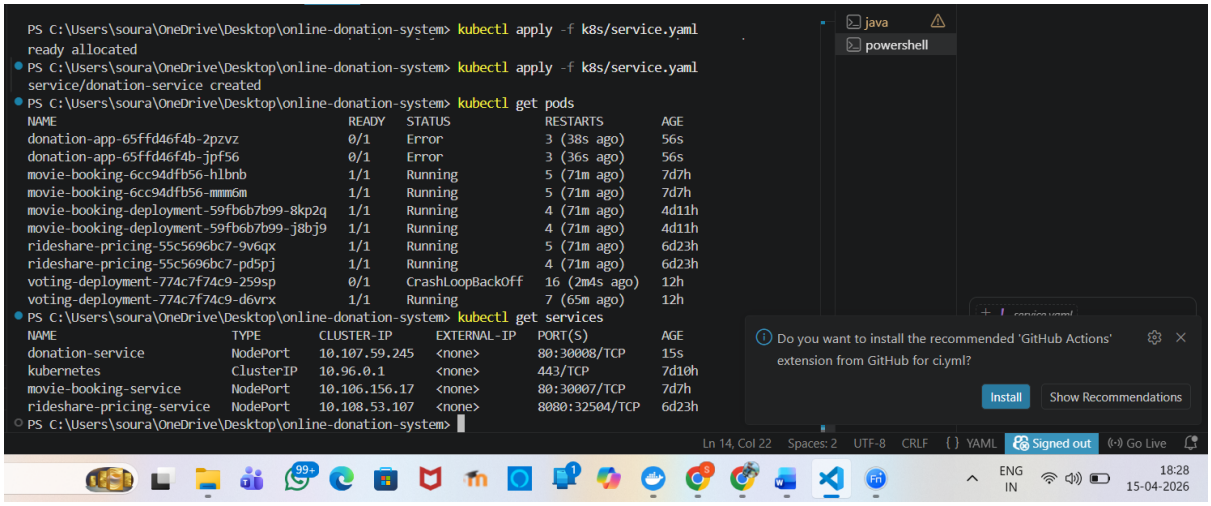
Docker:

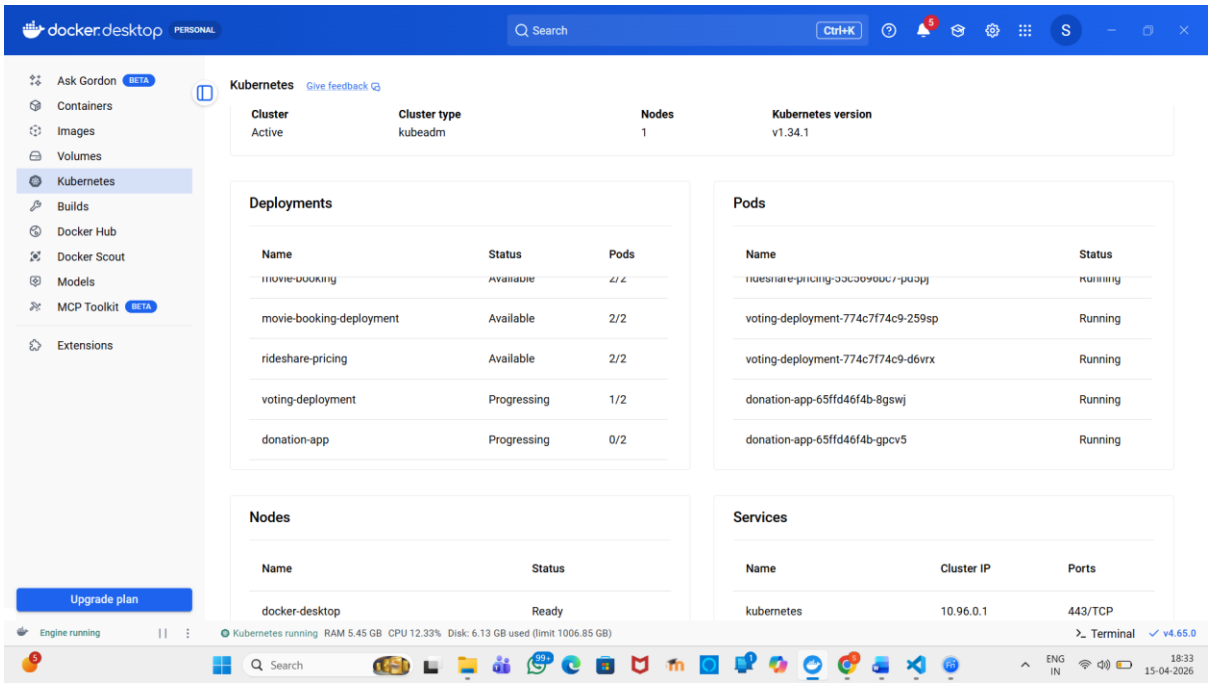




4.Deploy services on Docker and Kubernetes

Kubernetes:





YAML Files

Deployment.yaml

apiVersion: apps/v1

kind: Deployment

metadata:

  name: donation-app

spec:

  replicas: 2

  selector:

```
    matchLabels:
      app: donation

  template:
    metadata:
      labels:
        app: donation

    spec:
      containers:
      - name: donation-container
        image: donation-app
        imagePullPolicy: Never
        ports:
        - containerPort: 8080
```

Service.yaml

```
apiVersion: v1
kind: Service
metadata:
  name: donation-service

spec:
  type: NodePort
  selector:
    app: donation

  ports:
    - port: 80
      targetPort: 8080
      nodePort: 30008
```